MNIST and CIFAR-10: native-workflow timing comparison

20 October 2026

Non-IID, seed 42, 100 initial clients, 40% FR3. Our v15.4 data: 100 completed rounds per dataset. STD-DAGMM: 100 completed rounds per dataset, five fresh fitting epochs each, no removals. These totals are measured, not projected. Components absent from historical logs are not estimated. Network communication was not measured. Different native trajectories, hardware load, local attacker evaluation and timer boundaries preclude a causal speedup comparison. The CIFAR detector encountered near-zero available RAM, so paging may affect its timings; paging was not measured.

Table 1: Measured STD-DAGMM component times (seconds per round, mean ± sample SD, 100 rounds per dataset).

Component	MNIST	CIFAR-10
Client training	33.986 ± 1.539	76.208 ± 4.164
Simulated FR response	0.062 ± 0.006	0.255 ± 0.055
Server aggregation	0.042 ± 0.004	0.189 ± 0.043
Global evaluation	2.144 ± 0.073	4.547 ± 0.227
Update preparation	0.045 ± 0.004	0.127 ± 0.015
Detector fitting	13.629 ± 0.467	45.029 ± 1.671
Detector scoring	0.278 ± 0.011	0.856 ± 0.035
Flag decision	0.000 ± 0.000	0.027 ± 0.024
All detector work	13.952 ± 0.476	46.040 ± 1.695
Combined compute	50.186 ± 1.943	127.239 ± 5.478

Our historical training, aggregation, detection and trap-generation components: not separately recorded. Network transmission: not measured in either local simulator. STD has no traps. All detector work includes preparation, fitting, scoring and decisions; it overlaps those table rows and must not be added a second time.

Table 2: Native-workflow timing over 100 executed rounds per method and dataset. All entries are measured; this is not a controlled speedup experiment.

Dataset	Workflow / horizon	Rounds	Mean s/round	Sum (min)
MNIST	Ours, measured full horizon	100	34.41	57.35
MNIST	STD-DAGMM, measured full horizon	100	50.19	83.64
CIFAR-10	Ours, measured full horizon	100	84.85	141.41
CIFAR-10	STD-DAGMM, measured full horizon	100	127.24	212.06

Table 3: Our 100 measured rounds grouped by events. Within-category mean is not detector-only overhead; amortized contribution is category sum / 100.

Dataset	Round category	Count	Within-category mean (s)	Contribution (s/round)
MNIST	Warm-up	10	37.00	3.70
MNIST	Probe without cycle decision	27	34.29	9.26
MNIST	Full-cycle decision	3	34.45	1.03
MNIST	Single audit	6	33.95	2.04
MNIST	Ordinary	54	34.04	18.38
CIFAR-10	Warm-up	10	117.76	11.78
CIFAR-10	Probe without cycle decision	63	82.04	51.68
CIFAR-10	Full-cycle decision	7	84.26	5.90
CIFAR-10	Single audit	2	76.76	1.54
CIFAR-10	Ordinary	18	77.51	13.95

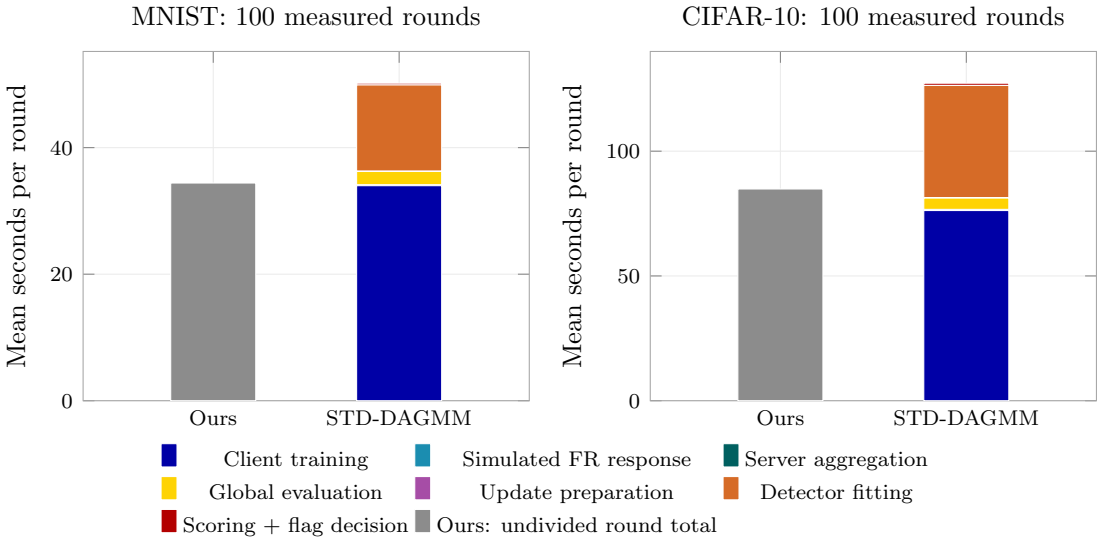


Figure 1: Mean timing over 100 executed rounds per method and dataset: non-IID FR3, seed 42, 100 initial clients, 40% free riders. STD-DAGMM has measured task components; our gray bars are undivided whole-round totals, not zero-cost detection. Both means include rounds 1–100. Different trajectories, timer coverage and host load prevent a causal speedup claim.

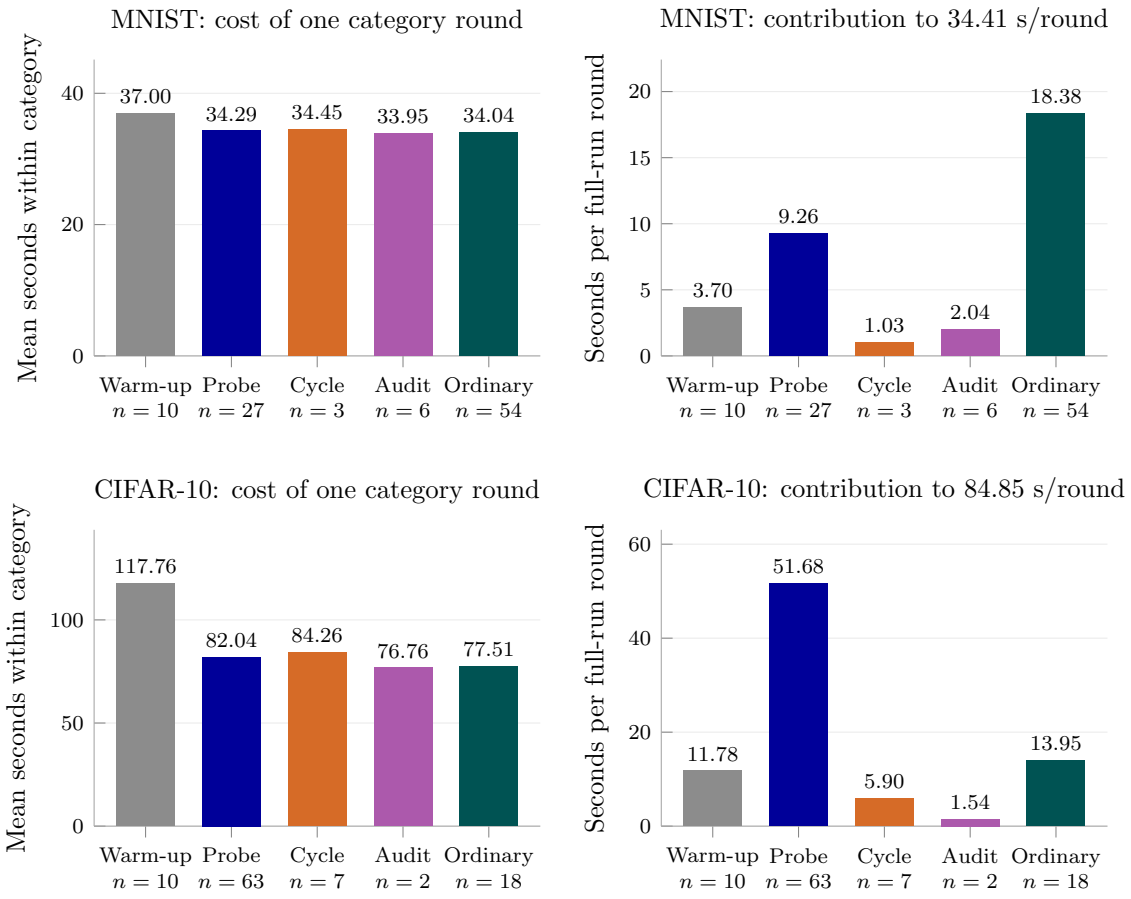


Figure 2: Where our 100-round runtime is spent (non-IID FR3, seed 42, 40% free riders). Rows separate MNIST and CIFAR-10. Left: mean duration of a complete round in each category. Right: summed duration for that category divided by 100; the five bars add to the full-run mean printed in the title. Labels give category counts n. Probe excludes cycle-decision rounds; Cycle includes the final cycle decision; Audit is a single audit round. Categories are mutually exclusive. A cycle bar includes client training, aggregation and evaluation: it is NOT detector-only time. Axes scale independently; all bars start at zero.

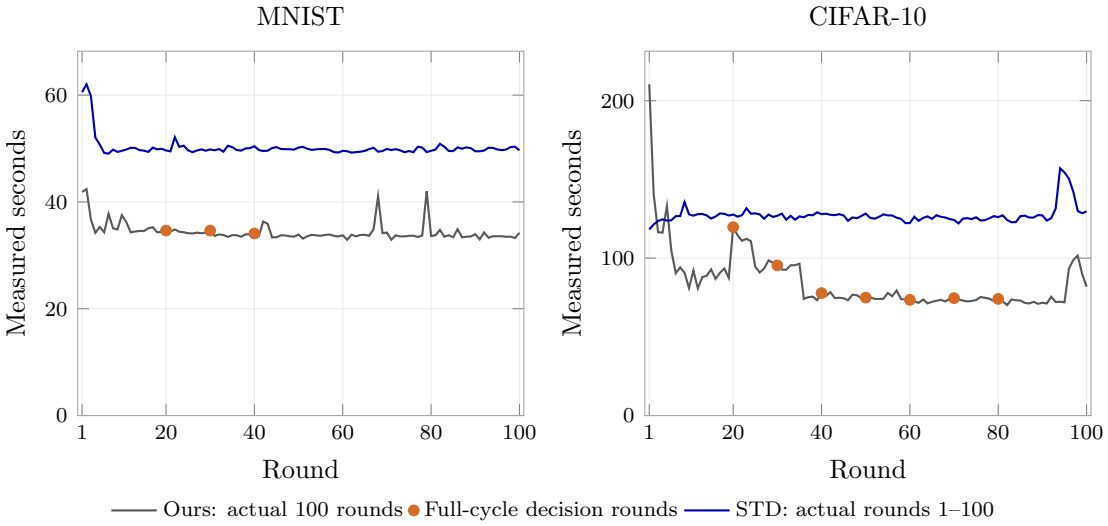


Figure 3: Actual per-round durations over 100 executed rounds for both methods. Our trajectories include changing active-client counts and probing schedules; STD-DAGMM retains all 100 clients. Markers indicate full-cycle decisions, not their isolated computational cost. Axis scales differ by dataset.